HILB^D IS LIMIT PRESERVING FOR FINITELY PRESENTED X

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 305-310 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 312-328, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_pic.json (pb=885, stmt 305-310, proof 312-328).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01ZP: sources/stacks-pic/pinned-original-source.tex, source lines 328–328. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06AC: sources/stacks-pic/pinned-original-source.tex, source lines 328–328. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZM: sources/stacks-pic/pinned-original-source.tex, source lines 328–328. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05LX: sources/stacks-pic/pinned-original-source.tex, source lines 328–328. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes. If $X \to S$ is of finite presentation, then the functor* $\mathrm{Hilb}^d_{X/S}$ *is limit preserving (Limits, Remark 05LX).*

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $T = \lim T_i$ be a limit of affine schemes over S. We have to show that $\mathrm{Hilb}^d_{X/S}(T) = \mathrm{colim}\, \mathrm{Hilb}^d_{X/S}(T_i)$. Observe that if $Z \to X_T$ is an element of $\mathrm{Hilb}^d_{X/S}(T)$, then $Z \to T$ is of finite presentation. Hence by Limits, Lemma 01ZM there exists an i, a scheme Z_i of finite presentation over T_i, and a morphism $Z_i \to X_{T_i}$ over T_i whose base change to T gives $Z \to X_T$. We apply Limits, Lemma 01ZP to see that we may assume $Z_i \to X_{T_i}$ is a closed immersion after increasing i. We apply Limits, Lemma 06AC to see that $Z_i \to T_i$ is finite locally free of degree d after possibly increasing i. Then $Z_i \in \mathrm{Hilb}^d_{X/S}(T_i)$ as desired. $\square$